Multenions and Differential Invariants.—II.

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§ 11. *The fundamental covariant vector linity,* η.—The meaning of a manifold as based on a quadratic differential form is here taken for granted, and this, whether, in the ordinary terminology, all or only some of its dimensions are real. Let

$$\left.\begin{array}{ll} \iota_a = i_a, & (a = 1, 2, \ldots, c), \\ \iota_b = i_b\sqrt{-1}, & (b = c+1, c+2, \ldots, n), \\ \rho = \sum\limits_{h=1}^{n} x_h \iota_h. & \end{array}\right\} \tag{1}$$

c having any given value from 1 to n. In the present paper we shall deal with multenions and multenion linities based on a system of vectors ρ for which all the co-ordinates x_h and the similar scalars are real. In the case of relativity $n = 4$, $c = 3$.

The fundamental quadratic form is taken to be

$$ds^2 = V_0 d\rho\eta d\rho = V_0 d\rho'\eta' d\rho', \tag{2}$$

the sign which is given to ds^2, a scalar, being convenient in applications to relativity, but inconvenient in applications to pure geometry. η, a self-conjugate vector linity, is a given function of position, ρ; ρ', a vector, an arbitrary function of ρ; and since the transformation from η to η' is given by (2) the fundamental linity η is said to be covariant. [η and ds^2 are real, but sometimes ds and sometimes $ds\sqrt{-1}$ is real; when ds is real it is "a time-like interval"; when $ds\sqrt{-1}$ is real it is a length; when the question of reality is left undecided ds is an interval.] At any given position ρ' may be so chosen that $\eta' d\rho' = d\rho'$, that is $\eta' = 1$. This imposes that η is without singularities; it has n mutually orthogonal principal directions and all its roots are real and positive, excluding zero values. Thus $\eta^{\frac{1}{2}}$ is assumed to have the same orthogonal system and its roots are taken to be the positive values of the square roots of η so that $\eta^{\frac{1}{2}}$ is real and without ambiguity.

We may, and from here on shall, take our previously introduced scalar k (§8) to be $|\eta^{\frac{1}{2}}|$, the determinant of $\eta^{\frac{1}{2}}$. [We have

$$\eta = \chi'\eta'\chi$$

so that

$$|\eta| = |\eta'| \times |\chi|^2 = |\eta'| \times (k/k')^2,$$

because k/k' was defined by

$$kdb = k'db' \qquad \text{or} \qquad k/k' = |\chi|. \quad]$$

It will be seen that when, in the usual notation of relativity, it is said that with Galilean co-ordinates

$$(g_{11}, g_{22}, g_{33}, g_{44}) = (-1, -1, -1, +1),$$

we have not that $\eta(\iota_1, \iota_2, \iota_3, \iota_4) = (-\iota_1, -\iota_2, -\iota_3, +\iota_4)$ but instead the simpler statement that $\eta = \eta^{\frac{1}{2}} = 1$ or

$$\eta(\iota_1, \iota_2, \iota_3, \iota_4) = (\iota_1, \iota_2, \iota_3, \iota_4),$$

for it will be observed that this gives

$$ds^2 = -dx_1^2 - dx_2^2 - dx_3^2 + dx_4^2.$$

A little care is required in expressing a linity, its conjugate, reciprocal, etc., by scalars or co-ordinates. It seems sufficient to illustrate from vector linities. When we say that ϕ is given by the square array of 16 scalars

$$\begin{vmatrix} a & b & c & d \\ e & f & g & h \\ l & m & n & p \\ q & r & s & t \end{vmatrix}$$

we imply that the co-ordinates of $\phi\iota_1$ are found in the first *column* thus

$$\phi\iota_1 = a\iota_1 + e\iota_2 + l\iota_3 + q\iota_4$$

and so for the other columns. In the square array of n^2 scalars specifying ϕ, let a_{xy} be the scalar in the xth row and yth column. Thus $\phi\iota_y = \sum_{x=1}^{n} a_{xy}\iota_x$, or more conveniently

$$a_{xy} = V_0 \iota_x^{-1} \phi \iota_y. \tag{3}$$

Except when $c = n$ the square array of a self-conjugate linity is not symmetrical about the principal diagonal. Let a_{xy}, a'_{xy} be the square arrays of ϕ and its conjugate ϕ'. By the meaning of conjugate we have

$$V_0 \iota_x \phi \iota_y = V_0 \iota_y \phi' \iota_x.$$

Thus instead of $a'_{yx} = a_{xy}$ (as we have when $c = n$) we have by (3)

$$a'_{yx} = \iota_x^2 \iota_y^2 a_{xy}, \tag{4}$$

and in particular, when ϕ is self-conjugate $a_{yx} = \iota_x^2 \iota_y^2 a_{xy}$. If we would retain the 16 scalars g_{ab} with their usual meanings as in treatises on relativity we may use $g_{(ab)}$ for the corresponding scalars specifying η. Thus from the fact that $ds^2 = \sum g_{ab} dx_a dx_b$ and from (3)

$$g_{ab} = V_0 \iota_a \eta \iota_b, \qquad g_{(ab)} = V_0 \iota_a^{-1} \eta \iota_b,$$

whence

$$g_{ab} = \iota_a^2 g_{(ab)}. \tag{5}$$

More details with regard to these notations and the theory of tensors will be found below in §17.

Note on Notations in the Present Paper.—There will be many exceptions below to the conventions here to be described. Especially do exceptions occur in very frequently recurring symbols. Thus none of the following receive the "cross" notation although they should do according to the conventions,

$$\lambda,\ E_a,\ \bar{E}_a,\ C_\omega,\ C.$$

λ is a covariant vector, and of the other four C_ω, C are contramixt linities while $E_a, \bar{E}_a$ are portions of contramixt linities. The explanations of the technical terms will be made as they come to be used below.

(1) Contravariant multenions will have no distinguishing mark. Covariant multenions will usually be distinguished by a "cross" (which, because of its position, is scarcely likely to be confused with a sign of multiplication), thus

$$\alpha^\times,\ u^\times,\ p^\times,\ q^\times,$$

read "alpha cross," etc.

(2) Covariant and comixt linities will have no distinguishing mark. Contravariant and contramixt linities will usually be distinguished by a cross.

(3) Contravariant and covariant linities will be denoted by Greek letters; contramixt and comixt linities by Roman capitals.

(4) $\alpha, \beta, \gamma, \alpha^\times, \beta^\times, \gamma^\times$, will generally denote "dummy" vectors. (Eddington uses "dummy" in an analogous sense for a positive integer.) Dummies are always (below, though not of necessity) treated as constants in that they are not affected by the differentiations of ∇.

The duty of a dummy contravariant or covariant multenion may be generally described as "to occupy, and thereby indicate, a situation into which an actual, previously defined, contravariant or covariant multenion, may legitimately be inserted."

§ 12. *The Invariant Forms of any Multenion Expression.*—η has been defined as a fundamental vector linity. It is also used for the corresponding extensive linity, a multenion linity; and similarly for $\eta^{\frac{1}{2}}$. Since in (2), of §11, $ds^2 = (\eta^{\frac{1}{2}} d\rho)^2$, $\eta^{\frac{1}{2}} d\rho$ in our manifold takes the place of the $d\rho$ of multenions in a Galilean manifold. [In a Euclidean manifold all the dimensions are real. In a Galilean manifold some are imaginary.] More generally, at a given position of the manifold, $\eta^{\frac{1}{2}} q$ where q is a contravariant multenion, or $\eta^{-\frac{1}{2}} q^\times$ where $q^\times$ is a covariant multenion, takes the place of q in a Galilean manifold. $\eta^{\frac{1}{2}} q$ and $\eta^{-\frac{1}{2}} q^\times$ will be spoken of as the neutral forms of q and $q^\times$. Indeed, the three q (contravariant), ηq (covariant) and $\eta^{\frac{1}{2}} q$ (neutral), are to be regarded as only different mathematical representations of the same intrinsic property, of the manifold; just as in an ordinary dynamical system, the whole system of velocities or instead the whole system of momenta equally will serve to specify the instantaneous motion.

Corresponding to any multenion expression such as $V_a \cdot pqV_b(rs)$ which has a Galilean interpretation there are always the three forms (p,q,r,s now being contravariant) in our manifold (contra., co., neut.).

$$\eta^{-\frac{1}{2}} V_a \cdot [\eta^{\frac{1}{2}} p \eta^{\frac{1}{2}} q V_b (\eta^{\frac{1}{2}} r \eta^{\frac{1}{2}} s)], \qquad \eta^{\frac{1}{2}} V_a \cdot [\,], \qquad V_a \cdot [\,].$$

The first and second of these three can *always* be expressed in such a shape that only η and not $\eta^{\frac{1}{2}}$ occurs explicitly; though the expression may be very complicated as in the example chosen. It is these final forms from which $\eta^{\frac{1}{2}}$ has been explicitly banished, that are regarded as invariantive.

To show this, first note as an example, that the product pqr can be first modified by expressing qr as a sum of terms such as $V_{a+b-2c} \cdot V_a q V_b r$ and then $p \cdot qr$ can be similarly modified. We have then merely to find the invariant forms (two, namely, covariant and contravariant) of $V_{a+b-2c}uv$. [The reader is reminded that in our notation u,v,w, have homogeneities a,b,c, respectively, or have the forms $u = V_a p,\ v = V_b q,\ w = V_c r$.]

Let a be greater or equal to b and let u,v be contravariant. Then most frequently all we require to note are the two cases $c = 0, c = b$:

$$\eta^{-\frac{1}{2}} V_{a+b} \eta^{\frac{1}{2}} u \eta^{\frac{1}{2}} v = V_{a+b} uv, \qquad \eta^{\frac{1}{2}} V_{a+b} \eta^{\frac{1}{2}} u \eta^{\frac{1}{2}} v = V_{a+b} \eta u \eta v,$$

$$\eta^{-\frac{1}{2}} V_{a-b} \eta^{\frac{1}{2}} u \eta^{\frac{1}{2}} v = V_{a-b} u \eta v, \qquad \eta^{\frac{1}{2}} V_{a-b} \eta^{\frac{1}{2}} u \eta^{\frac{1}{2}} v = V_{a-b} \eta uv.$$

and here of course we may have $u = \eta^{-1} u^{\times}$ or $v = \eta^{-1} v^{\times}$ in each of the forms.

To deal with the general value of c in $V_{a+b-2c}uv$ we use ζ. Now we know that if $\tau_1, \tau_2, \ldots \tau_n$ are n independent contravariant vectors at a point, then their normal reciprocals $\hat{\tau}_1, \hat{\tau}_2, \ldots \hat{\tau}_n$ are n independent covariant vectors. Also we know that if $(\alpha, \beta^{\times})$ is any expression—bilinear in two vectors

$$(\zeta, \zeta) = -\sum_{a=1}^{n} (\tau_a, \hat{\tau}_a).$$

Hence if we insert ζ, ζ into two places in an invariant form, of which one is occupied by a contravariant vector and the other by a covariant vector, the resulting expression will be invariantive (meaning, "of invariant form"). Now it is easy to see by reduction to primitive units that

$$V_{a+b-2c} uv = [(-1)^{\frac{1}{2}c(c+1)}/c!] V_{a+b-2c} \cdot V_{a-c} u \zeta^{(c)} V_{b-c} \zeta^{(c)} v, \tag{1}$$

where as usual $\zeta^{(c)}$ stands for $\zeta_1 \zeta_2 \ldots \zeta_c$ and $\zeta_c^{(c)}$ for $V_c \zeta^{(c)}$. Hence the contravariant form of $V_{a+b-2c}uv$ is

$$[(-1)^{\frac{1}{2}c(c+1)}/c!] V_{a+b-2c} \cdot V_{a-c} u \zeta_c^{(c)} V_{b-c} \zeta_c^{(c)} v, \tag{2}$$

from which the covariant form may be written down, and either $u^{\times} = \eta u$ or $v^{\times} = \eta v$ or both may be used in place of u and v. In my MS. work I am in the habit of using $\zeta_K^{(c)}, \zeta_Q^{(c)}$ for $K\zeta_c^{(c)}, Q\zeta_c^{(c)}$. Thus (2) would be $(c!)^{-1}V_{a+b-2c}\cdot V_{a-c}u\zeta_c^{(c)}V_{b-c}\eta\zeta_K^{(c)}v$. A special case of (2) often required is that the contravariant form of $V_2\omega\omega'$ where ω and ω' are contravariant and ${}_nV_2$ is

$$V_2(V_1\zeta\omega)(V_1\eta\zeta\omega'). \tag{3}$$

or more generally, for infinitesimal rotations generally, the contravariant form of $V_a\omega\iota$ is

$$V_a(V_1\zeta\omega)(V_{a-1}\eta\zeta\iota). \tag{4}$$

§ 13. *Normal and Incident Components in Invariant Form.*—These components occur very frequently in relativity. Thus the energy-momentum linity is density multiplied by an incident component, at any rate in its kinetic or primitive form. Normal and incident components serve to separate: (1) vector potential from electrostatic potential; (2) momentum from energy; (3) force from power; (4) current density from charge density; (5) magnetic induction from electrostatic force; (6) magnetic field strength from displacement. One should always be prepared to recognise these resolutions.

In quaternions if α is a given vector and σ an arbitrary one, σ has two components with reference to α, namely the normal component

$$N_0\sigma = \alpha^{-1}V\alpha\sigma = \alpha V\alpha\sigma/\alpha^2$$

and the incident component

$$I_0\sigma = \alpha^{-1}S\alpha\sigma = \alpha S\alpha\sigma/\alpha^2.$$

As will be familiar to users of the quaternion method, it is these components each multiplied by α^2, rather than the components themselves, which we shall expect to meet with frequently in analytical expressions. The two

$$N_\alpha\sigma = N\sigma = \alpha V\alpha\sigma, \qquad I_\alpha\sigma = I\sigma = \alpha S\alpha\sigma,$$

may be referred to as the normal and incident *expressions* of σ with reference to α. With regard to these four linities N_0, I_0, N, I we have as follows

$$\left.\begin{array}{lll} N_0^2 = N_0, & I_0^2 = I_0, & N_0I_0 = 0 = I_0N_0, \\ N_0' = N_0, & I_0' = I_0, & N_0 + I_0 = 1 = N_0' + I_0' \end{array}\right\}, \tag{1}$$

$$\left.\begin{array}{lll} N^2 = \alpha^2N, & I^2 = \alpha^2I, & NI = 0 = IN, \\ N' = N, & I' = I, & N + I = \alpha^2 = N' + I' \end{array}\right\}. \tag{2}$$

Similarly in a Galilean manifold any multenion q has two components N_0q, I_0q normal and incident to a given vector α. Let

$$q = \sum_{c=1}^{n} w, \qquad \text{where} \quad w = V_cq;$$

then when the linities Nw, N_0w, etc., have been defined, Nq, N_0q have also been defined. We have

$$\left.\begin{aligned} Nw &= \alpha V_{c+1}\alpha w, & Iw &= \alpha V_{c-1}\alpha w, \\ N_0 w &= \alpha^{-1}V_{c+1}\alpha w, & I_0 w &= \alpha^{-1}V_{c-1}\alpha w \end{aligned}\right\}. \tag{3}$$

[If ε is a vectorium of homogeneity b, a vector σ has a normal component $\varepsilon^{-1}V_{b+1}\varepsilon\sigma$ and an incident component $\varepsilon^{-1}V_{b-1}\varepsilon\sigma$ with reference to ε. With reference to ε, w has many, not merely two, components given by

$$w = \varepsilon^{-1}V_{b+c}\varepsilon w + \varepsilon^{-1}V_{b+c-2}\varepsilon w + \cdots.$$

The terms on the right of this equation are all, separately, but not obviously, of homogeneity c.]

Next let α be a given contravariant vector at a given position ρ of our manifold and q an arbitrary contravariant multenion at the same position and let $w = V_c q$. Also let α_v, q_v, w_v be the neutral forms and $\alpha^\times, q^\times, w^\times$ the covariant forms of α, q, w respectively, that is to say,

$$\left.\begin{aligned} \eta^{\frac{1}{2}}\alpha &= \alpha_v = \eta^{-\frac{1}{2}}\alpha^\times, \\ \eta^{\frac{1}{2}}w &= w_v = \eta^{-\frac{1}{2}}w^\times, \\ \eta^{\frac{1}{2}}q &= q_v = \eta^{-\frac{1}{2}}q^\times \end{aligned}\right\}. \tag{4}$$

The invariant form of α^2 is α_v^2 or

$$V_0\alpha\eta\alpha = \alpha_v^2 = V_0\alpha^\times\eta^{-1}\alpha^\times. \tag{5}$$

With our present meanings (3) is not invariant in form. There are four modes of choosing the invariant forms of N and I, N_0 and I_0 and generally of any multenion linities, (1) covariant, (2) contravariant, (3) comixt, (4) contramixt. We will take number (3) of these for the basis of the notation in connection with N and I. A comixt linity means one which acting on a covariant multenion produces a covariant multenion. For this type of linity it is obvious that the invariant forms of (3) are given by

$$\left.\begin{aligned} Nw^\times &= V_c\alpha V_{c+1}\eta\alpha w^\times = V_c\eta^{-1}\alpha^\times V_{c+1}\alpha^\times w^\times, \\ Iw^\times &= V_c\eta\alpha V_{c-1}\alpha w^\times = V_c\alpha^\times V_{c-1}\eta^{-1}\alpha^\times w^\times, \\ N_0 w^\times &= Nw^\times/V_0\alpha\eta\alpha = Nw^\times/V_0\alpha^\times\eta^{-1}\alpha^\times, \\ I_0 w^\times &= Iw^\times/V_0\alpha\eta\alpha = Iw^\times/V_0\alpha^\times\eta^{-1}\alpha^\times \end{aligned}\right\}. \tag{6}$$

(2) comes with this notation to be modified in two ways only; α^2 is replaced by $V_0\alpha\eta\alpha$ or $V_0\alpha^\times\eta^{-1}\alpha^\times$ and in place of N' and I' we have a new kind of conjugate $\eta N'\eta^{-1}, \eta I'\eta^{-1}$ involving the fundamental linity η. Thus we now have

$$\left.\begin{aligned} N^2 &= V_0\alpha\eta\alpha\cdot N, & I^2 &= V_0\alpha\eta\alpha\cdot I, & NI &= 0 = IN, \\ \eta N'\eta^{-1} &= N, & \eta I'\eta^{-1} &= I, & N + I &= V_0\alpha\eta\alpha = \eta(N' + I')\eta^{-1} \end{aligned}\right\}. \tag{7}$$